

9 Does Stretching Affect Performance?

Static and PNF Stretching

At least from the mid-20th century, stretching was believed to enhance performance. An early study by the renowned George Dintiman reported that stretch training enhanced sprint speed (45). Higher baseline levels of flexibility in volleyball players were associated with better performance in agility, acceleration speed, and lower body muscular power as compared to players with low flexibility level (67). But a variety of studies have indicated no beneficial effects of stretch training on the oxygen cost of running (three days/week for ten weeks) (110), drop jumps (four days/week for six weeks) (153), vertical jumps (four days/week for six weeks) (8), or sprint speed (8). However static stretching two days/week for 8 weeks did improve rebound bench press performance (145). But the acute effect of static stretching within a pre-event warm-up has received far more attention lately.

It was in the 1990s that more frequent investigations began into the role of stretching as part of a warm-up prior to training or competition. The common thought since the 1960s was that static stretching would improve performance and Worrell and colleagues in 1994 (147) did report that four hamstring stretches of 15–20 seconds each improved eccentric and concentric leg flexion torque, as expected! As mentioned previously, the acute increase in range of motion (ROM) was thought to reduce resistance to movement, and provide greater limb excursions that would be important for work output (work = force x linear or angular distance), running stride length, activities that need substantial flexibility (e.g., gymnastics, dance, figure skating, and others) and other movements. The proposition that improved flexibility would improve performance arises from the concepts of stress, strain, stiffness and the modulus of elasticity. Stress and strain are related as they are defined either as the applied force divided by the tissue cross-sectional area (stress) or the ratio or change in tissue length with an applied force (strain). Stress and strain are directly related to tissue stiffness, as stiffness is defined as the ratio of stress to strain or the ratio of changes in force to changes in length. Often the term compliance is used as the opposite to stiffness. A more compliant muscle or tendon is a less stiff muscle or tendon. Historically, Robert Hooke

found that there was a proportional relationship between force and elongation. According to his modulus of elasticity, stiffer tissues need a greater stress (force or load) to produce a greater strain (change in length). Thus, a more compliant or less stiff musculotendinous unit should need less force or load to move it through a ROM resulting in a greater economy of movement and less resistance to movement.

But Kokkonen and colleagues in 1998 were among the first to contradict the prevailing notion that increased flexibility would improve performance. They found that five stretches involving six repetitions of 15 seconds each decreased knee flexion and extension force by 7–8 percent (92). A 1999 study by Avela et al. (3) had subjects receive one hour of repeated passive static stretching reporting impairments in force, electromyography (EMG) and reflex sensitivity. Typical of the problem with many of these earlier studies is the real-world relevance. What percentage of the population stretches their calf muscles for 60 minutes?

This led to many more studies in the early part of the millennium reporting stretching-induced impairments. A number of comprehensive reviews have reported that prolonged static stretching can lead to subsequent impairments in strength, power, sprint speed, agility, balance, evoked contractile properties and other performance and physiological parameters (12, 15, 87, 127). These impairments on average range from 4–7.5 percent. Static stretching can even impair the extent of muscle hypertrophy, strength and insulin-like growth factor-1 (IGF-1) (23) achieved if you consistently stretch before or during resistance training. These reports have led to changes in how athletes prepare or warm-up for training and competition. Whereas static stretching was widespread prior to the new millennium, there has been a shift away from static stretching to more dynamic stretching and dynamic activities. A review of American college tennis coaches in a 2012 article showed that 87 percent of coaches always employed a warm-up and within that warm-up, 38.5 percent used a combination of static and dynamic stretching, 10.5 percent only used static stretching and the rest relied primarily on dynamic activities and dynamic stretching (84). The results were slightly different for American university cross country and track and field coaches in a 2013 article with 98.1 percent always ensuring a warm-up and within that group 44.7 percent used a combination of static and dynamic stretching whereas 41.5 percent only performed dynamic stretching and 10.5 percent used static or proprioceptive neuromuscular facilitation (PNF) stretching (86). Their throwing counterparts used a warm-up 95.6 percent of the time with 40.7 percent using dynamic stretching, 38.5 percent employing a combination of static and dynamic stretching and 11.1 percent using static or PNF stretching (85). A French survey of 3,546 individuals (47.3 percent women and 52.7 percent men, national/international level athletes: 25.2 percent, regional level athletes: 29.8 percent, or recreationally active: 44.9 percent) reported that stretching was generally performed either after training (72.4 percent) or as a warm-up (49.9 percent), with static stretching primarily utilized (88.2 percent) except for the

warm-up when most people used dynamic stretching (86.2 percent) (4). So, while the warm-up paradigm has shifted to an emphasis on dynamic stretching or a combination, about 10 percent of coaches in these studies still use static stretching alone. Is this paradigm shift a legitimate movement based on irrefutable fact or one of the many fads that we see come and go over the decades?

With static stretching-induced impairments, the emphasis should be placed on the term “prolonged” static stretching. It is generally agreed that 60 seconds or more of static stretching per muscle group is more likely to cause significant performance decrements (12, 15, 87). Muscle length is also a mitigating factor. Static stretches performed at short muscle lengths showed substantially greater deficits (10.2 percent), while static stretching at long muscle lengths actually resulted in small strength gains (2.2 percent) (12).

PNF stretching results from our review (11 PNF studies) reported approximately 4 percent performance deficits (12). An earlier review (2007) found no clear evidence for a negative effect of PNF stretching on subsequent exercise performance (121). Sarah Marek and colleagues found that both static and PNF stretching led to similar strength, power, and muscle activation impairments but the magnitude of the deficits were small (104). Therefore, as usual, there is never complete agreement in science. There are actually some studies that show PNF-induced improvements in isokinetic torque (147) and postural stability (123). But contradicting those studies are others reporting either no changes in strength or jump height (151) or PNF-induced deficits in proprioception (132), movement time, hip flexion angular velocity (101), vertical jump height (24, 33), strength, power and muscle activation (104). One study showed that ten repetitions of PNF stretching increased blood lactate concentration indicating there would be a greater reliance on anaerobic metabolism following PNF stretching (69). In the few studies that compared static and PNF stretching, PNF presented greater (6.4 percent vs. 2.3 percent) subsequent deficits than the static stretching protocols. An important note to keep in mind is that most of these studies intervened only with prolonged stretching without the additional activity you typically experience in a warm-up. The methodological problems and misinterpretations of these types of studies will be examined later. But if you only static or PNF stretch and do not have time to commit to a full warm-up procedure, how important are the consequences?

While a 5 percent deficit (15) might not seem that dramatic to the average athlete, it can be devastating to an elite athlete. Usain Bolt won the 100-metre gold medal at the 2012 Olympic Games in 9.63 seconds. If a competitor was 5 percent slower, then where would he place in the competition. In fact, Gerard Phiri from Zambia was almost exactly 5 percent slower and ran the 100 metres in 10.11 seconds. Did Phiri medal? Did Phiri make it to the finals? No, Phiri came in second last place (15th) in the semi-finals. The 5 percent difference was devastating. Bolt makes millions of dollars from competing and endorsements, and I can bet that 99.9 percent of the readers have never heard of Phiri. Adidas, Nike, or Reebok are not knocking at his door throwing

money at him. Perhaps you watched the 2016 Olympics and saw Bolt win again in 9.81 seconds (he was getting older and slowed down!). As a Canadian, I was cheering for Andre De Grasse. He won a bronze medal in 9.91 seconds (see Figure 9.1). The difference between Bolt and De Grasse was 1.009 percent. Performing the correct warm-up, is unbelievably crucial in elite competition. In our 2011 review (15) we stated, “It would be wise to be cautious when implementing static stretching of any duration or for any population when high-speed, rapid stretch-shortening cycle, explosive or reactive forces are necessary, particularly if any decreases in performance, however small, would be important.” However, there are a number of caveats to this statement. We also mentioned that “All individuals should include static stretching in their overall fitness and wellness activities for the health and functional benefits associated with increased ROM and musculotendinous compliance.” In our 2016 review (12), which involved a more expansive and critical examination of the literature, we modified our views and said the following, “When a typical pre-event warm-up is complete (i.e., initial aerobic activity, stretching component, 5–15 minutes of task- or activity-specific dynamic activities), the benefits of SS and PNF stretching for augmented ROM and reduced injury risk balance, or may outweigh, any possible cost of (trivial) performance decrements.” Sounds very confusing! Should we stretch or not before explosive or high-speed training or competition?

To simplify the answer, it might be easier to think of yourself either as a Ferrari or a Cadillac (see Figure 9.2). Elite athletes are the Ferrari, Lamborghini or Bugati sport cars of the world. The purpose of the elite athlete is to push themselves to their physiological, anatomical and psychological limits.



Figure 9.1 Olympic sprint final 2016



Figure 9.2 Ferrari (tighter suspension) versus Cadillac (more compliant, less stiff suspension)

They need to produce maximal forces, or power, and many times in the briefest possible periods exerting extremely high pressures, torques, impacts and other stresses to their body. Usain Bolt and the other international-calibre sprinters have foot contact times on the track of less than 100 milliseconds. Thus, they must hit the ground and take-off without absorbing and losing much energy. Elite sprinters will leave the starting blocks in about 120–160 milliseconds (107). The fastest start was by Bruny Surin (another Canadian!) at 101 milliseconds in the 1999 world championship semi-final. Athletes with stiffer or less compliant musculotendinous systems are reported to be more economical runners (less energy expended for the same amount of work) than their more flexible counterparts (59, 83, 140). A number of studies have reported that static stretching can reduce sprinting speed (9, 31, 55, 125, 128, 146). Similar to human sprinters, a Ferrari race car needs to accelerate from the start and move out of curves and corners rapidly. To accomplish this task, race cars have tight suspension so they react to the road immediately. Driving a Ferrari over a bumpy road would not be much fun and in fact excruciatingly painful for your lower back as the suspension would not absorb much of the shock of the pot holes in the roads. For the average city or country road you want a car with nice soft suspension that absorbs the shocks so that the driver feels almost nothing when encountering a bump in the road. A Cadillac and other similar cars have great suspensions that effectively absorb these shocks. You would never expect a Cadillac to beat a Ferrari in a race especially on a route that included curves and turns. After every race, mechanics must examine, repair or replace the damaged components of race cars. There are always damaged parts after every race. The car has been pushed to its mechanical limit and the suspension did not adequately absorb or reduce the effect of the road environment on the other parts of the Ferrari. You would understandably be very upset, if every time you drove your Cadillac you had to bring it to the mechanic to get it repaired. The top sprinters (Bolt and colleagues), tennis players (Roger Federer, Novak Djokovic, and colleagues) and other individual sport athletes as well as team sport athletes have athletic trainers, physiotherapists, massage therapists and

others ready to provide them with treatment immediately after a competition when they have pushed their bodies to the limit. The average fitness enthusiast and weekend warrior athlete does not have a cadre of health practitioners to treat him/her after every workout. The majority of the population does not push themselves to the extreme limit.

There is no single correct prescription for the optimal degree of flexibility. Everybody has their “Goldilocks” zone. The elite sprinter needs to transfer contractile forces as quickly as possible from the muscle to the tendon to the bone resulting in movement. If a tendon is overly slack (high compliance) there will be a negative impact on electromechanical delay (EMD). The EMD is the time it takes for the muscle to be activated, contract, pull and tighten the tendon and then have the bone move. Under laboratory conditions the EMD is measured as the duration between the onset of the muscle action potential wave (M-wave) to the onset of force or torque production (see Figure 9.3). If an elite athlete’s parallel (muscle membranes) and series elastic (i.e., tendons) components were very compliant then the EMD is prolonged and the time it takes for a muscle contraction to actually result in movement is also lengthened. This of course is anathema to the elite sprinter who wants to explode out of the blocks in the shortest time (reaction time) or spend the least amount of time in contact with the track (increased stride rate).

Previously, I told you that prolonged static stretching or increased flexibility can decrease running economy. But there are also studies that contradict these previous studies with reports that static stretching also decreased the energy cost of running (60) or had no effect (72). How is this possible? Were one set of scientists incompetent and incorrect? No, it is necessary to look at the specificity of

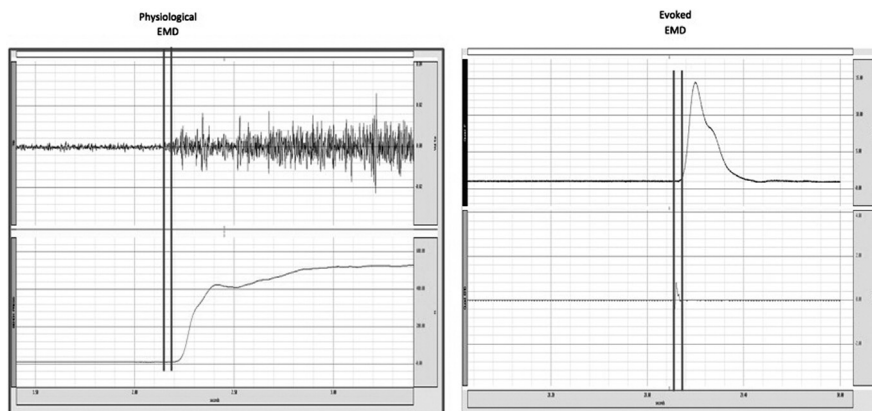


Figure 9.3 Electromechanical delay (EMD). Physiological EMD is the time from the onset of muscle activation (illustrated with electromyography in the first figure) and the onset of voluntary force. Evoked EMD is the time from the onset of evoked activation (illustrated by the electrically evoked muscle action potential in the second figure) to the onset of evoked twitch force. EMD in these examples are approximately 40–60 milliseconds

the task. Schmidtbleicher (126) defined two types of rebound jumps or stretch shortening cycle (SSC) actions (see Figures 9.4 and 9.5). A jump performed with a contact time of less than 250 milliseconds was classified as a short SSC and more than 250 milliseconds was considered a long SSC. The two studies that found improved running economy with prior stretching used submaximal, slower, prolonged running (i.e., 40, 60, 80 percent of VO₂ maximum or below lactate threshold) with long SSC. With slower running and longer SSC, the person is in longer contact with the ground (see Figure 9.6). A very stiff musculotendinous system absorbs the reaction forces over a very short duration and returns the energy to the muscles very quickly (desirable for sprinters). If the system is very stiff, the mechanical and reflex energy return (stretch-shortening cycle) would occur while the slower runner is still in the landing phase and not ready to propel themselves into the next stride. A more compliant or flexible muscle and tendon would absorb the reaction forces over a longer period and the spring-like action of the muscles and tendons would return the energy to the muscles when the distance runner is ready to push off the ground.

Studies that examine evoked contractile properties (electrical stimulation of the muscle or nerve) might give some insight into some of the mechanisms underlying static stretch-induced impairments. A few studies have reported static stretch-induced decreases in twitch force, rate of force development and prolonged EMD (29, 36, 52, 122). As a reminder, EMD is primarily affected by the musculotendinous unit (MTU) compliance or stiffness. It is suggested that there is a direct relationship between EMD and muscle stiffness (63). If the MTU is more compliant it will take longer to transfer the energy of the myofilament cross-bridge kinetics to the muscle and tendon tissues to move the bone. Muscle stiffness has a greater influence on athletic performance

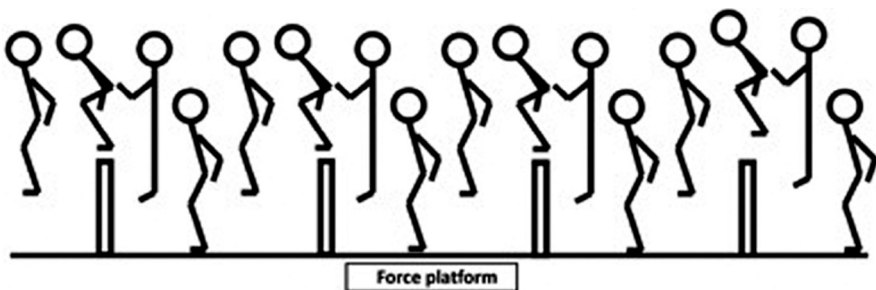


Figure 9.4 Stretch-shortening cycle: Hurdle jumps (from right to left) beginning with take-off (propulsion), flight phase, (clearance of the hurdle), landing (involving eccentric contractions to absorb the reaction forces), with a short contact (amortization or transition) time, followed by concentric contractions for another propulsion phase. The eccentric contractions during landing stiffen the musculotendinous unit (MTU) to decrease electromechanical delay while elongation of the MTU activates muscle spindle stretch receptors (nuclear bag and chain fibres) to induce reflex contractions to augment the force of the subsequent concentric contractions



Figure 9.5 Stretch-shortening cycle: Drop jump preparation, landing (amortization or transition phase with elongation/stretching of muscles and tendons) and take-off (propulsion: concentric contractions)

than flexibility (120). The intrinsic or baseline stiffness of a muscle can still be moderately high and yet with a high tolerance to stretch (pain or discomfort), an individual can still have substantial levels of flexibility.

Some researchers have compared a mechanomyogram signal to the force or EMG signal to parcel out the EMD associated with excitation contraction coupling and MTU viscoelastic properties. A mechanomyogram (MMG) monitors changes in the muscle shape by sensing vibrations and oscillations of the muscle fibres at the resonant frequency of the muscle. With a higher

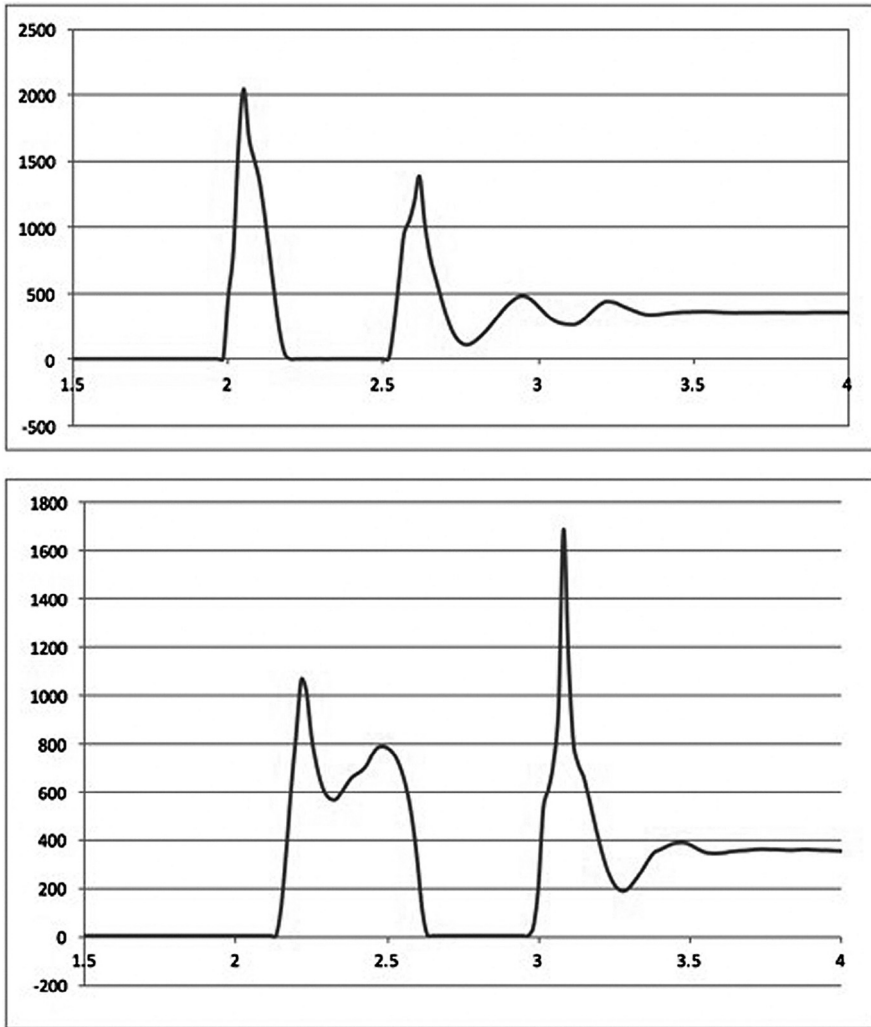


Figure 9.6 Drop jump stretch-shortening cycle reaction forces with a short versus long contact time. The first peak is the landing from a drop jump (contact time), followed by the flight phase (second baseline) and the last peak is the landing following the flight phase

signal-to-noise ratio than the surface EMG it is purported to be able to monitor muscle activity from deeper muscles. After five sets of 45 seconds of passive stretching, the excitation-contraction (E-C) coupling aspect of the EMD (Δt : EMG-MMG) was prolonged for approximately 15 minutes, whereas the MTU visco-elasticity (Δt : MMG-tetanic force) was protracted for up to two hours (51). It is necessary to remind the reader that research is

almost never unanimous. A study from our lab also found force deficits up to two hours after stretching two muscles for three sets of 45 seconds each. However, while there were significant voluntary force deficits (10 percent), the evoked twitch and tetanic force deficits (1–4 percent) were non-significant over that same period (114). However, peak tetanic force has also been reported to be depressed for two hours after five repetitions of 45-second static stretches with 15 seconds' rest between stretches (52). Changes in evoked twitch forces represent changes in E-C coupling (muscle action potential leading to the release and sequestration of calcium from the sarcoplasmic reticulum) and tetanic forces represent changes in myofilament kinetics. Hence, five x 45 seconds of stretching in this one study damaged the myosin and actin cross bridges, which would affect not only peak force but likely the rate of force development.

The Goldilocks flexibility or ROM zone for sprinters versus distance runners would be different based on their differing ground contact times. Further evidence for task specificity of stretching is provided by experiments that demonstrated a 5 percent increase in rebound chest press (145) and no significant effects on isometric bench press (136). As the amortization or transition phase from the eccentric to the concentric phases of a bench press would be much slower than the contact time during sprinting, a more compliant system would be able to absorb and return the elastic energy in a more appropriate task specific time period.

Effects of Static Stretching on the Stretch-Shortening Cycle

The stretch-shortening cycle (SSC) is an important neuromuscular mechanism that enhances power production with rapid, explosive activities and contractions. Prime examples of SSC activities are plyometric activities such as countermovement, hurdle and drop jumps, hopping, bounding, skipping, sprinting, and other similar activities. With SSC activities, external forces (e. g., body mass accelerated by gravity) lengthen the muscle during an active contraction (eccentric phase), which is followed by a concentric (shortening) contraction (93, 112). SSC exercises use the elastic energy that was stored during the eccentric contraction phase to augment the concentric phase (28, 94). This augmentation involves both mechanical (musculotendinous) and neural (reflexive) components. Like an elastic band, the contractile elements (i.e., myosin) and muscle connective tissue (i.e., titin, desmin) and tendons are stretched resulting in the storage of potential elastic energy that is returned to the neuromuscular system shortly thereafter. The energy stored in the system can be extremely high. For example, a study from our laboratory reported that hurdle jumps that emphasized short contact times generated mean reaction forces of 4,335 Newtons (442 kg or 972 lbs). This reaction force was approximately 5.5 times the body mass and approximately 100 percent higher than the much slower SSC action of countermovement jumps (27). The elongation of the MTU does not need to be extensive to achieve this elastic

rebound. A tendon elongation of 6–7 percent during the stretching phase is sufficient to elicit the SSC. Rack and Westbury (118) originally labelled this small elongation: short-range elastic stiffness. If the tendon can return elastic energy with only a 6–7 percent elongation, we can see why sprinters may not need or want a very compliant MTU. A stiffer MTU is still fine for transferring elastic recoil energy.

The duration for the return of this elastic recoil is dependent upon the compliance or elasticity of the SSC. A tighter system will return this elastic recoil earlier than a more compliant system. Slow SSC actions only get a minor contribution from the passive elastic elements, creating a greater reliance on muscle contractile properties to produce concentric phase force. However, a rapid SSC can take greater advantage of elastic energy from the MTU (39, 80). With rapid SSC actions, the cross bridges can remain attached during the stretch, to augment the force during the concentric phase (6). The rapid deceleration during the eccentric stretch phase followed by the acceleration during the propulsive, concentric, phase contributes to a power potentiation and improved energy economy that is dependent on the speed of the eccentric stretch phase (80).

Not only is there a mechanical elastic storage and energy return but the SSC causes an excitation of the muscle spindle reflexes (111). Rapid SSC actions induce greater reaction forces, muscle reflex activity and muscle activation (137). The muscle spindles sense and react to changes in the extent and rate of muscle lengthening. Within the muscle spindle, nuclear chain fibres preferentially respond to changes in the amount of stretch or elongation of the muscle while the nuclear bag fibres respond to both the extent of elongation and the rate. These signals return to the spinal motoneurons via annulospiral (high conduction velocity) and flower spray endings (low conduction velocity) through Ia (nuclear bag and chains) and II (nuclear chains) afferents respectively. A monosynaptic myotactic reflex is initiated from the rapid stretch action resulting in the depolarization/activation (contraction) of the alpha motoneuron of the stretched muscle and inhibition (di-synaptic) of the antagonist muscle (105). This reflex action aids in neural potentiation, stiffening of the contractile components due to the myofilament crossbridge attachments and augmenting the active contraction forces.

The spindle afferents can contribute up to 30 percent to the activation of the motoneurons (57). Persistent inward currents (PIC) from prior contractions contribute to motoneuron depolarization during the subsequent contractions. The PIC is a depolarizing inward current of motoneurons that respond to brief synaptic input with prolonged firing activity, even after the input stops. (known as self-sustained firing). They can also increase the maximum motoneuron firing rate. PICs are derived from Na^+ and low voltage-activated L-type Ca^{2+} ion channels. A small afferent signal that activates the dendritic PIC is amplified creating a larger current to the soma (motoneuronal body) that produce higher motoneuronal firing rates

(50). It is interesting that neurophysiologists use the term “warm-up” for the increased activation of the Ca^{++} PIC. Thus, the increased reflex afferent activity from repeated SSC actions in concert with supraspinal drive would potentiate the motoneurons response (i.e., greater rate coding). However, a number of static stretching studies have reported H-reflex (afferent excitability of the spinal motoneuron) inhibition with stretching (3, 53, 64, 65, 66, 133). In contrast, work from the Blazeovich lab by Pulverenti et al. reported a lack of motor evoked potential (MEP) (115, 117) or H-reflex amplitudes (117) changes following stretching in the soleus (five x 60 seconds constant torque static stretches) during submaximal contractions, suggesting that changes in the excitability of the corticospinal-motoneuronal (115, 117) and Ia afferent spinal reflex (117) pathways do not contribute to post-stretch neural impairments. So, is there cortical, spinal, PIC, or afferent excitability inhibition or not with stretching? Two years after the Pulverenti publication, the Trajano and Blazeovich team published a narrative review on the topic (138). They divulged that there is clear evidence for prolonged static stretch-induced neural drive reduction and their conclusion was that the reduction in motoneuron excitability caused by a reduction in PIC strength was the most likely causative factor. PIC reduction was dependent on reduced facilitation by monoaminergic neuromodulatory systems (i.e., noradrenaline). Hence, static stretching moves the sympathetic-parasympathetic balance towards a more parasympathetic relaxation, negatively influencing motoneurons’ ability to maximally fire resulting in lower maximal muscle force (138). These conclusions are in accord with a prior narrative review by the Trajano, Nosaka, and Blazeovich crew (139) when they highlighted that primary factors inducing prolonged static stretch-induced force impairments are related to reductions in the central nervous system motor command as well as a decreased amplification of the spinal motor command.

There are many explosive, power type athletes such as gymnasts and figure skaters who have extensive flexibility and yet still have a rapid and efficient SSC. Does this not contradict this previous discussion? These types of athletes can perform powerfully even with a very compliant MTU if they actively stiffen the system before contracting. Hence it is important for them to pre-activate the system. More experienced and accomplished athletes will pre-activate the muscles with a feedforward (pre-planned) central command (44). The first 40 milliseconds of ground contact time is too soon for spindle reflexes to stiffen the system and thus pre-activation from supraspinal activation is necessary (61).

Stretching Effects on Balance

While we often connect the importance of improved balance to athletic performance, we have to remember that slips, trips, and falls contribute to 95 percent of hip fractures in seniors (73) and are the most common cause of

traumatic brain injury (81). There is some evidence that improved dorsiflexion ROM can facilitate balance test performance (anterior direction in the Star Excursion Balance Test: SEBT) (77) and even with 10–30 minutes of static or dynamic stretching (15-second repetitions) with a five-minute aerobic warm-up there was a small magnitude improvement in SEBT performance (19). The SEBT is usually performed under controlled and slower conditions and thus a less stiff, more compliant system that reacts over a more prolonged period could provide better feedback and postural adjustments for this test. The improved SEBT performance with greater flexibility might not necessarily translate to the far more rapid adjustments to posture (metastability) encountered with such high velocity perturbations found with downhill skiing, rugby, American football, and other sports or everyday activities like slipping on an icy sidewalk. Our narrative review (18) reported on balance impairments with static stretching durations of 30 seconds up to five minutes (six studies) contrasting with seven studies that found balance improvements with durations ranging from two repetitions of 15 seconds to 30 minutes of static stretching. There were three dynamic stretching studies which reported balance improvements (19, 32, 54)

Once again, we have to differentiate between the effects of prolonged static stretching and briefer durations of stretching. Prolonged static stretching has been shown to adversely affect the spinocerebellar pathways (five x 60 seconds of stretching) (116), and ankle proprioception (six x 40 seconds) (130), whereas other studies using three x 30 seconds of static stretching showed no effect (97) or even improvements (58) on knee position sense (proprioception).

In addition to the importance of the vestibular and proprioceptive afferent and efferent

responses for balance (1, 10) the musculotendinous system needs to react and overcome disruptive torques or perturbation in order to restore the centre of gravity to within the base of support (metastability, 89). While a more compliant musculotendinous system may absorb perturbations over a longer duration, allowing greater sensory (afferent) feedback and efferent postural adjustments, a system that is too slack (compliant) necessitates greater shortening of the muscle prolonging the electromechanical delay (time to take up the slack of a compliant tendon in order to exert force on a bone for movement) (18). Hence, the system may react too slowly.

But you also need to differentiate between passive and active stiffness. While prolonged stretching-induced decreases in passive musculotendinous unit are reported, active musculotendinous stiffness may not concomitantly be reduced (18). Thus, while the initial absorption of a perturbation (e.g., a slip or trip) may be more reliant on passive stiffness/compliance, the reaction with an active muscular contraction would be much stiffer accelerating the muscular response. A good example would be the extreme passive compliance of a gymnast or a figure skater to achieve extraordinary feats of flexibility and balance but they can still explode off the floor or ice with jumps, flips, and tumbles due to the active contractions of agonist and antagonist muscles

(active stiffness). Furthermore, as stretch training can increase force output at longer muscle lengths, an individual losing their balance may move a leg further to expand the base of support while reacting more forcefully to counterbalance the perturbation (18).

Effects of Flexibility Training

Are the effects of stretching mitigated by population age, sex, trained state or other considerations? Perhaps the reason that so many studies report static stretch-induced deficits is due to the fact that most people do not stretch on a regular basis. If we took a group of untrained people and had them resistance train or go for a long run, they would certainly be sore and weak for days afterwards. Untrained people would experience deficits even with mild to moderate strength or aerobic training. So, would people who train for stretching regularly not experience these stretch-induced deficits? A study from our laboratory reported that 5 weeks of stretch training improved ROM by 12–20 percent but the participants still experienced deficits of 6–8 percent in leg strength and jump height (13). In the same study, we tested a diverse group of people and ranked them by their flexibility. We thought that perhaps the most flexible people might not be as adversely affected by prolonged static stretching. We performed a cross-sectional correlation and did not find any significant relationship between hip and ankle ROM with static stretch-induced deficits in force and jump height. Furthermore, the Behm and Chaouachi review (15) examined 41 stretching studies with trained individuals versus 68 studies that use untrained individuals. They did not find any significant difference overall between the two groups with both experiencing stretch-induced deficits. Prolonged static stretching shows no favourites, trained and untrained, flexible or inflexible people are still susceptible to the adverse effects.

Ageing Effects

There are very few studies examining the effects of stretching on middle aged and older populations. The reports are contradictory with some studies reporting no impairments (71) and others showing impairments (17). Thus, it seems that while the results from individual studies can be contradictory, in general all segments of the population from young to old may experience static-stretch induced impairments from prolonged static stretching that is not accompanied by the dynamic active components of a full warm-up.

Effect of Stretching Intensity

The literature emphasizes the importance of stretch duration for determining whether performance deficits will be incurred (12, 15, 87). These reviews agree that less than 60 seconds of static stretching per muscle group should be

implemented if static stretch-induced impairments are to be minimized. What about the intensity of stretching? If some individual stretches to the point of maximal discomfort or pain is it more likely to lead to performance deficits than if they performed lower intensity stretching? Is it “more pain, less gain” for performance after stretching? Once again, the answer is not clear from individual studies. There are some studies showing no impairments when the stretch was held to a point of “mild” discomfort (90, 91, 103, 150), which are contradicted by others that do show performance decrements with stretching to a point of mild discomfort (24, 37, 38, 78, 125). In a study from our laboratory, participants stretched the quadriceps, hamstrings and plantar flexors for 3 repetitions of 30 seconds each either to 100 percent, 75 percent or 50 percent of point of maximal discomfort. All stretching intensities adversely affected jump height. To be cautious, individuals should expect performance deficits if they stretch at any intensity for more than 60 seconds per muscle group without the inclusion of any other dynamic activity within the warm-up.

Cost Benefit Analysis

To consider how much flexibility is needed by an individual, a subjective cost-benefit analysis should be conducted. The possible **COST** of prolonged static stretching without a full warm-up is performance impairments of 4–8 percent. The possible **BENEFITS** of static stretching are a greater ROM, more shock absorption, and lower incidence of musculotendinous injuries. A stiff musculotendinous system does not absorb forces well and is more susceptible to strains and sprains. Furthermore, how much ROM is necessary to achieve your goals. For a sprinter, soccer rugby, or ice hockey player, and other sportspeople, a moderate degree of flexibility is necessary. A gymnast, dancer, figure skater, competitive diver, ice hockey goalie, and others need more extreme ROM. To the vast majority of the population who are recreationally active, the cost of possible minor impairments pales in comparison to the cost of injury. A reasonable individual would probably surmise that hitting the golf ball ten yards or metres shorter is a reasonable price to pay to ensure that he/she can continue playing golf multiple times a week throughout the season without injury.

Effects of Dynamic Stretching on Performance

In contrast to static and PNF stretching, dynamic stretching is reported to produce (mean of 48 studies) trivial to small (1.3 percent) performance enhancements (12). The Canadian Society for Exercise Physiology position stand and meta-analysis illustrated that dynamic stretching improved performance in 20 studies (small or greater than small effect sizes), showed trivial effects in 21 studies and impairments in seven studies (12). Measures included countermovement jumps, sprints, agility, isometric and isokinetic force or torque and power, 1RM, balance and other variables. Another review by two

French researchers, Opplert and Babault (113) indicated that there is substantial evidence highlighting the positive effects of dynamic stretching on ROM and subsequent performance (force, power, sprint and jump). They suggested that the greater ROM could be primarily attributed to muscle-tendon unit stiffness reductions, whereas enhanced performance was accredited to elevated temperature and post-activation potentiation enhancement-related mechanisms related to the voluntary contractions of the dynamic stretching.

The average reader should always be cognizant that even when a textbook or a review reports overall significant or meaningful improvements or impairments, a close inspection of the individual studies in the literature almost always shows a spectrum of results that are influenced by the protocol, subject/participant population, the type of measures or other variables. Thus, general summaries or recommendations can be made on the global review of the literature, but there is always the possibility for each individual that their response to a stressor like stretching might not behave similarly to the grand aggregate means reported in these books and reviews.

The review by Behm and Chaouachi (15) recommended that more 90 seconds of dynamic stretching should be performed for each muscle group, but the studies are quite variable in their durations, so a dose response relationship for dynamic stretching is more difficult to establish (12). In these studies, the durations of dynamic stretching can range from one to 20 minutes with either some performance enhancement or no adverse effects. Similarly, there seems to be a tendency for performance improvements with faster dynamic stretches or more intense ballistic stretches, but there is a high degree of variability in the literature, so definitive statements regarding dynamic stretching duration or frequency (speed) also cannot be made at this time. According to one study, the dynamic stretching should be performed before the general warm-up (low-intensity aerobic speeds for an average of five minutes at an individual pace, followed by two x 10 metres sprint) to achieve the best sprint times (70). Another review by our very productive Japanese colleagues (148) recommended that dynamic stretching be conducted “as fast as possible”. They also found that explosive performance may be impaired as the dynamic stretching volume increases and thus their regression analysis indicated that the optimal repetitions or distance and sets was “10–15 repetitions” or “10 yards–20 metres” x “1–2 sets”, respectively (148). So, if dynamic activities and stretching tend to excite the central nervous system (CNS) whereas static stretching tends to depress the CNS, would a combination of dynamic and static stretching within a warm-up counterbalance each other?

Effects of Combining Static and Dynamic Stretching on Performance

As reported earlier, the studies by Judge and colleagues reported that 39–45 percent of American college coaches for tennis, and track and field use a combination of static and dynamic stretching in their warm-up. Unfortunately,

the real-world practices are not paralleled by the volume of research in this area. Thus, even with the scant research combining dynamic and static stretching, can we say whether the neural excitation of the dynamic stretching counteracts the neural inhibition/disfacilitation of the static stretching? Not always! The research results vary from impairments to no change to improvements. In one study, the combination of static and dynamic stretching resulted in 50-metre sprint time impairments compared to dynamic stretching alone (55). Another study showed no difference and no vertical jump impairments when comparing dynamic stretching to a combination of static and dynamic stretching (142). But maybe Gaelic footballers are unique, as in a study using Irish athletes, 20 and 40-metre sprint speed (0.7–1.0 percent) and counter-movement jump height (8.7 percent) improved with a combination of static and dynamic stretching (100). A study from our lab, used five conditions 1) general aerobic warm-up with static stretching; 2) general aerobic warmup with dynamic stretching; 3) general and sports specific warm-up with static stretching; and 4) general and sports specific warm-up with dynamic stretching. We found that when a sport specific warm-up was included, there was a 1 percent improvement ($p = 0.001$) in 20-metre sprint time with both the dynamic and static stretch groups. Without a sports specific warm-up, there were no such improvements. In addition, static stretching increased sit and reach (hamstrings and lower back) ROM approximately 3 percent more than the dynamic stretching. In another study, the combination of static and dynamic stretching was not detrimental for the agility of professional soccer players, however dynamic stretching was more effective for preparing for agility (88).

Is performance affected if the individual stretches between sets of a resistance training workout? A study that compared heavy load resistance training (three sets x four repetitions) with heavy load resistance training with stretching (three x 15 seconds) between sets reported no effect of inter-set stretching on jump height (62).

As the research recommends less than 60 seconds of static stretching per muscle group so as not to induce impairments and that many of the ROM studies favour static stretching over dynamic stretching for the greatest ROM increases, it would seem reasonable to recommend a combination of brief (≤ 60 seconds) static stretching with more than 90 seconds per muscle group of dynamic stretching in addition to a prior $\geq$ five minutes of aerobic activity and a subsequent 5–15 minutes of activity specific dynamic activity to ensure an optimal ROM with either no deficits or perhaps even enhancement of performance. This type of warm-up has been reported in the literature to increase ROM without subsequent impairments or in some cases actually improve some aspects of performance (12).

Effects of Ballistic Stretching on Performance

Ballistic and dynamic stretching are often confused as being the same action. As mentioned before, dynamic stretching involves moving a joint actively

through a full ROM under controlled conditions (not maximum speed), whereas ballistic stretching is usually performed at higher speeds often with bouncing movements at the end of the ROM. As usual, the literature is not unanimous. Some articles report that ballistic stretching has no significant effect on subsequent strength (5, 7), vertical jump height (24, 82) or plantar flexors passive resistive torque (102), while others report decreases in knee flexion and extension strength (98, 109), fatigue endurance (7, 98), passive resistive torque and muscle stiffness (95) and Achilles tendon stiffness (102). Fletcher used the term fast dynamic stretching for movements at 100 beats per minute, which might be considered ballistic. With these fast dynamic (maybe ballistic) stretches, they found an increase in vertical jump height and EMG compared to slow dynamic stretches or no stretching at all. Konrad et al.'s (95) multivariate analysis indicated no clinically relevant difference between acute bouts of static, ballistic and PNF stretching. In all these studies, a full warm-up (initial aerobic component, static stretching or slow dynamic stretching followed by sport specific dynamic activity) was not incorporated. Hence, these cited studies should warn us about performing ballistic stretches as the only activity in a warm-up as there is some evidence for performance decrements. The following section describes the problems of interpreting such delimited studies when applying their results to the real world.

Full Warm-ups

The emphasis in the previous section was to warn about the negative effects of "PROLONGED" static and PNF stretching. The typical stretch durations for professional basketball, American football, ice hockey, and baseball players ranges from 12–17 seconds (47, 48, 49, 129). Powerlifters surveyed from 51 countries did static stretches before training for an average of 6.72 ± 10.31 stretch repetitions for an average duration of 30.8 ± 31.4 seconds. Dynamic stretching was performed by 90.4 percent of the powerlifters (131). However, in the research you see stretch durations of 9–13 repetitions of 135 seconds each (20–30 minutes for the plantar flexors) (36, 56). One of our early studies had participants perform 20 minutes of quadriceps stretching (14). Scientists would say that these studies lack "ecological validity". That is, they are not representative of what actually happens in the real world. Almost nobody would stretch one muscle group for 20–30 minutes especially without all the components of a full warm-up before a training session or competition. Many of the static stretching studies in the literature lack ecological validity (real-world application). As you would imagine most of these prolonged static stretching studies had substantial stretch-induced deficits.

Another major problem is that most studies intervened with a stretching only protocol with no or little other dynamic activity (12). The traditional or conventional pre-activity warm-up protocol consists of an initial submaximal or low intensity exercise component (e.g., running, cycling) that increases the

heart rate and muscle temperature (remember the effects of heat on ROM) as well as exciting/activating the neuromuscular system. This aerobic segment is followed by static stretching, which according to our recommendations should not exceed more than 60 seconds per muscle group (12, 15, 87). Dynamic stretching should also be involved to provide more task and velocity specific ROM augmenting activity. The final aspect involves activity or sport specific dynamic activity that further activates the cardiovascular, neuromuscular, endocrine and other physiological systems in preparation for the activity (12, 149). The few studies that do include a full warm-up typically do not illustrate significant stretch-inducing deficits or may provide some performance improvements (99, 108, 124, 134, 143). One of our collaborative studies demonstrated that the combination of prior running, followed by static stretching finishing with countermovement jumps resulted in 7–17 percent better jump performances versus static stretching alone (152). In another collaborative study with our colleagues in Tunisia, the combination of static and dynamic stretching at different static stretching intensities (point of discomfort and less than point of discomfort) resulted in no significant impairments. Thus, the inhibition or depression of the neuromuscular system with prolonged static stretching seems to be balanced by the excitation of the system with dynamic warm-up activities.

A fairly definitive study by our lab (119) demonstrated relatively clearly the interaction of static stretch duration and the dynamic components of the warm-up. In this study, participants went through a full warm-up protocol including a 5-minute aerobic warm-up, followed by a static stretching component, dynamic stretching component, and finally a sport specific dynamic activity component. They either did static stretches of 30, 60, or 120 seconds or no static stretching within this warm-up protocol. Participants were tested after each component as well as ten minutes after the warm-up. All conditions improved hamstrings and quadriceps ROM. There were no impairments in MVC forces, jump height with the control and 30 seconds of static stretching at any time. Impairments were evident with 120 seconds of stretching for quadriceps force (post-static stretching and at ten minutes' post warm-up), force produced in the first 100 milliseconds (post-static stretching), and muscle activation (ten minutes post-warm-up). Vertical jump improved with the warm-up activities except for 60 and 120 seconds of static stretching when tested immediately after the static stretching component. Thus, as the Behm et al. review (12) proposed, static stretching of less than 60 seconds per muscle group within a full warm-up should not lead to performance impairments. The lack of static stretch-induced impairments when incorporated within a full warm-up could be due to the post-activation potentiation effect of the dynamic activities. Kummel et al. (96) reported the typical static stretch deficits but when they added hops, there was no difference from the control condition but the jump heights were significantly higher than the stretch only condition. Similarly, ten minutes of dynamic stretching were sufficient to potentiate vertical jump characteristics (141). Even adding a deadlift did not

augment the potentiating effect on vertical jump. Thus, dynamic stretching alone can potentiate performance and offset possible impairments from prolonged static stretching.

These findings were also substantiated by Blazeovich and colleagues (21) who incorporated 30 seconds of static stretching into a full warm-up and reported increased ROM but no impairments in jumping, sprinting or agility tests. A very interesting aspect of their study was that they asked the subjects about their performance expectations when stretching was included. Before the study, 18 out of 20 participants nominated dynamic stretching as the most likely to improve performance and 15 out of 20 participants nominated no stretching to be least likely. Nonetheless, these ratings were not related to test performances. The authors suggested that including static or dynamic stretching into a warm-up routine instilled more confidence in the participants regarding their performance in the ensuing sports-related tests. As psychological effects are so potent for optimal performance, and appropriate durations of static and dynamic stretching generally do not seem to degrade performance when incorporated into a full warm-up, it might be a good idea to include short to moderate durations of static stretching (<60 seconds) into a complete warm-up.

Limitations of Acute Stretching Studies

So, athletes, coaches, fitness instructors and enthusiasts have been cautioned by a number of reviews since 2011 about the negative effects of prolonged static stretching especially without comprehensive dynamic warm-up components (12, 15, 30, 87). We published an updated review in 2021 and found 27 acute studies published since the 2016 review, that reported on average a mean moderate magnitude (8.04 percent; $d = 0.55$) ROM increase and a small magnitude (−1.5 percent; $d = 0.36$) performance impairment after static stretching (16). Unfortunately, there were still ecological validity problems with only 18 out of 27 (66.6 percent) studies including a full warm-up, 18 out

Table 9.1 Full warm-up components

<i>Submaximal Aerobic Activity</i>	<i>Static Stretching</i>	<i>Dynamic Stretching</i>	<i>Task-Specific Activities</i>
5–15 minutes of running, cycling or other activity to increase muscle temperatures, decrease tissue visco-elasticity, increase heart rate, enzymatic cycling and other factors	<60 seconds per muscle group No need to go to the point of discomfort or pain Stretch major muscle groups and specific muscle groups to the activity	>90 seconds per muscle group Use full range of motion with a controlled movement at moderate speeds	5–15 minutes Practice movements that are associated with the sport or task at velocities close to the actual movement

of 27 (66.6 percent) also implementing more than 60 seconds of static stretching, four out of 25 (14.8 percent) added dynamic stretches or activities after static stretching and only ten out of 27 (37.1 percent) performed testing at more than five minutes after stretching.

After the warm-up, do most athletes or teams immediately begin their competition? Many athletes or teams go back to their dressing room or bench to finalize tactics, strategies, make equipment adjustments, and other necessities (i.e., pre-game renal voiding). For some competitions, there is a national anthem to start the game. But the average time to testing after a warm-up was 3.2, 4.9, and 4.1 minutes for static, dynamic and PNF stretching studies (12). Once again, the ecological validity can be questioned, as many athletes would not start their competition less than 5 minutes after their warm-up.

Furthermore, who are the subjects in these studies? As the majority of the studies are conducted at the Kinesiology, Physical Education, Sport or Exercise Science departments of universities, the students are likely to be familiar with these studies that report stretch-induced impairments. A study from our laboratory tried to deceive the participants by telling non-Kinesiology students that stretching would actually improve their performance and compared their responses to Kinesiology students (knowledgeable about the stretching literature) after 3 static hamstrings stretches of 30 seconds each. Both groups still experienced hamstring force and activation deficits due to stretching but the deceived group experienced improvements in the performance of the antagonist quadriceps muscle. Therefore, the deception could not overcome the stretching impairments of the hamstrings (90 seconds was 50 percent more than the recommended ≤ 60 seconds) but the quadriceps, which were not stretched, were susceptible to the deception that stretching would improve performance. Hence bias and deception could play a small role in these types of studies. Furthermore, there is bias in the journal publications for studies that provide significant findings (46). For example, it is more difficult to publish a study that reports no changes with stretching than a study that would report significant decrements. Our review found that reporting of data/statistics within published papers was less common when non-significant results were obtained. In most studies the details of non-significant changes were commonly not reported, which could result in an overestimation of the stretching deficit effects (12).

Improving Performance by Limiting Muscle Soreness

Excessive unaccustomed activity especially those activities that emphasize eccentric contractions can lead to exercise-induced muscle damage (EIMD) leading to delayed onset muscle soreness (DOMS) that can persist for 5–7 days (11, 68). EIMD and DOMS can decrease force, power, running speed, neuromuscular efficiency (extent of muscle activation needed to accomplish a task), ROM and many other performance factors (26, 34, 35). It has been a common belief that stretching can either prevent or reduce EIMD and

DOMS. Stretching either before, during or after the EIMD-induced exercise did not reduce subsequent DOMS (25, 68, 75, 76). A review by Howatson and van Someren (79) also concluded that stretching to prevent or reduce EIMD provides only minimal effects in reducing muscle soreness with no performance effects. Some early studies reported that during DOMS, static stretch-induced reductions in muscle activity (measured with EMG or measurement of the H-reflex to muscle action potential ratio) (41, 42, 43, 135) contributed to reduced soreness. For a similar reason (decrease muscle activity) (43, 74), static stretching tends to relieve muscle cramps (20, 40, 144). Other possible suggested mechanisms for DOMS relief would be that stretching helps to remove some of the inflammation or swelling associated with DOMS (22). Furthermore, stretching during DOMS could initiate the Gate Control Theory of Pain (106) suppression whereby the activation of higher velocity afferents (i.e. type Ia, Ib,) from mechanical stimulation of stretch, cutaneous and other receptors can inhibit the slower velocity pain afferents such as type IV afferents (2) (see Figure 9.7). A common example of the Gate Control Theory is the typical response to an injured finger or limb or other body part. The immediate response is to grab that injured body part and start rubbing it. Rubbing the area activates the high velocity cutaneous, tactile or pressure receptors that help to peripherally and centrally block the slower afferents from the nociceptors (pain receptors). Typically, there is never just one mechanism and thus DOMS reduction could be a combination of a number of mechanisms.

Summary

Throughout the second half of the 20th century, it was commonly believed that static stretching-induced increases in muscle compliance (reduced stiffness) would enhance subsequent performance by allowing a greater ROM with less resistance to movement. However, static and PNF stretching studies from the late 1990s into the early 2000s predominantly reported subsequent performance impairments of 3–7.5 percent. These findings changed the traditional warm-up paradigm shifting the emphasis from static to dynamic stretching within a warm-up. Specifically, it was found that prolonged static stretching (>60 seconds per muscle group) without all components of a full warm-up increased the chances for performance deficits. The decision to include static stretching within a warm-up may depend upon the type or demands of the activity (i.e., jogging with minimal ROM needs versus hurdling with increased flexibility needs), the level of the athlete (elite vs. recreational) and the priority of reducing the incidence of musculotendinous injuries compared to possible 3–4 percent performance decrements. Prolonged static stretching without a full warm-up can impair performance in both trained and untrained individuals, young and old, whether the stretch is performed at a moderate or high intensity.

Dynamic stretching, for its part, may have either no significant impairments or actually provide small enhancements of subsequent performance.

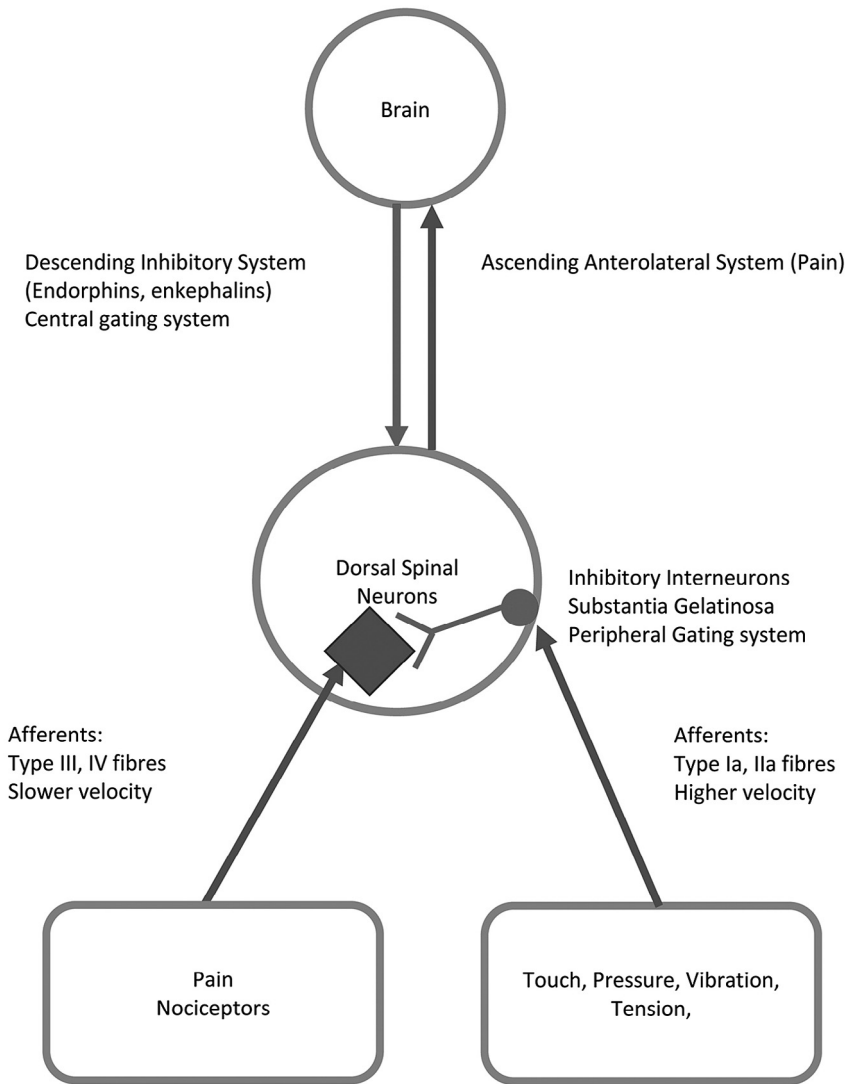


Figure 9.7 Gate Control Theory of Pain (Melzack and Wall)

Adding static to dynamic stretching tends to counterbalance possible impairments with prolonged static stretching. Individuals should incorporate at least 90 seconds of dynamic stretching per muscle group within a full warm-up. The full warm-up should include a 5–15-minute submaximal aerobic component, less than 60 seconds of static stretching per muscle group, more than 90 seconds of dynamic stretching per muscle group and finish with dynamic activity that should be activity or sport specific.

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